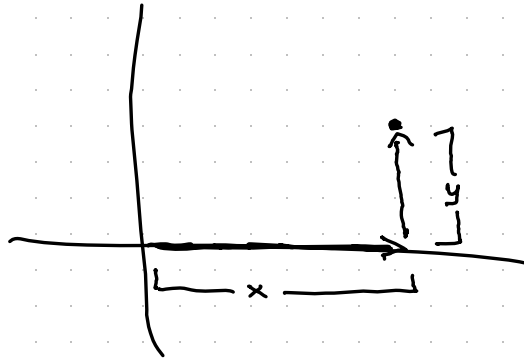
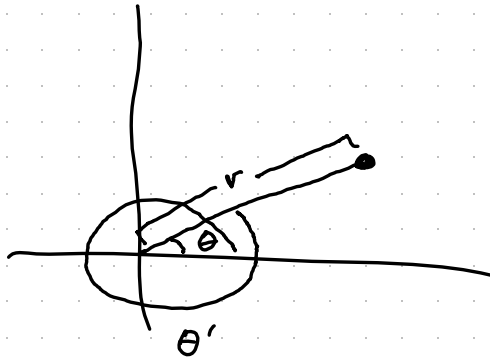


Polar coordinates — 7.3



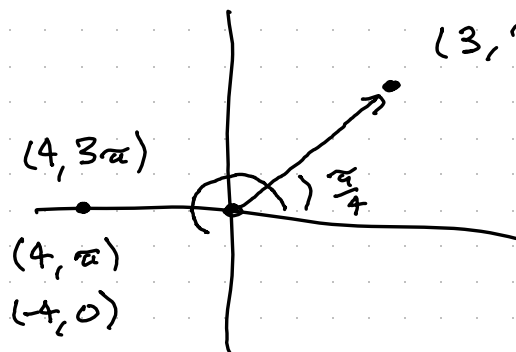
Cartesian coordinates:

tell you how to get to a point in the plane by moving horizontally and then vertically.



Polar coordinates:

tell you what direction to move in and how far to move



$$(3, \frac{\pi}{4}) = (3, 2\pi + \frac{\pi}{4}) = (3, -2\pi + \frac{\pi}{4})$$

$$= (3, \frac{\pi}{4} + 2\pi k)$$

for k an integer

$$= (-3, \pi + \frac{\pi}{4})$$

Polar coordinates (r, θ) are not unique.

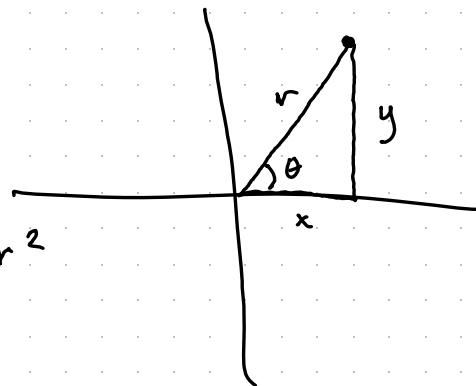
Converting back and forth:

$$\sin \theta = \frac{y}{r}, \quad \cos \theta = \frac{x}{r}, \quad \tan \theta = \frac{y}{x}$$

$$y = r \sin \theta$$

$$x = r \cos \theta$$

$$x^2 + y^2 = r^2$$



$$r = \sqrt{x^2 + y^2}$$

$$\theta = \arctan\left(\frac{y}{x}\right)$$

Examples: Convert the Cartesian $(5, 2)$
into polar coordinates.

$$r = \sqrt{25 + 4} = \sqrt{29}$$

$$\theta = \arctan\left(\frac{2}{5}\right)$$

$$(\sqrt{29}, \arctan(\frac{2}{5}))$$

Convert the polar $(6, \frac{\pi}{3})$ to Cartesian coords.

$$x = 6 \cos\left(\frac{\pi}{3}\right) = 6 \cdot \frac{1}{2} = 3$$

$$y = 6 \sin\left(\frac{\pi}{3}\right) = 6 \cdot \frac{\sqrt{3}}{2} = 3\sqrt{3}$$

$$(3, 3\sqrt{3})$$